Analysis of hydroacoustic ship signatures in multipath environments by cepstral F statistics

Andreas Galka, Alexandra Schäfke und Arne Schulz
Bundeswehr Technical Centre for Ships and Naval Weapons, Maritime Technology and Research (WTD 71), 24148 Kiel, Germany; E-Mail: andreasgalka@bundeswehr.org

Abstract

In this paper we study an aspect related to the analysis of hydroacoustic signatures of surface ships, namely the generation of spurious components due to multipath propagation of water sound in shallow water. We suggest to analyse data sets affected by multipath propagation by a variant of cepstral analysis which embeds the estimation of the cepstrum into a statistical F test. As an application example we analyse a real data set recorded in the Atlantic Ocean, close to the Portuguese coast, at a water depth of around 90 metres. We demonstrate that the cepstral F statistic can detect the delay times of low-order multipaths. Furthermore we show the typical shape of the time-varying cepstral F statistic, i.e., the cepstrogram.

Introduction

It is well known that when a ship passes a fixed hydrophone in shallow water, reflections at the water surface and the sea floor will generate a characteristic interference pattern in the spectrogram, sometimes called *range-frequency striations* [3]; if only reflections at the surface contribute, i.e., in deep water, the interference pattern is known as *Lloyd's mirror effect.* When analysing the hydroacoustic signature of ships, such interferences need to be taken into account, since they will produce spurious peaks in the power spectrum. This paper deals with the investigation of multipath propagation by cepstral analysis.

Time domain models

Assume a set of time series has been recorded by an array of N hydrophones, at a sampling rate of f_s. The data recorded by the ith hydrophone shall be denoted by $y_i(t)$ where $t = 1, \ldots, T$ denotes time in units of $1/f_s$. The time series $y_i(t)$ may be analysed in frequency domain, i.e., by applying the *Discrete Fourier Transform* (DFT); in this case *non-parametric* estimates of the power spectrum and the spectrogram would result. Alternatively, the time series may be analysed in time domain, i.e., by fitting predictive autoregressive (AR) models; in this case *parametric* estimates of the power spectrum and the spectrogram would result.

A well-known class of predictive autoregressive models is given by *linear state space models*, consisting of a pair of equations: a *dynamical equation* and an *observation equation.* The dynamical equation models the evolution of a time-dependent latent state variable $s_i(t)$, which is not directly observable; it would typically be given by a linear AR model, such as

$$s_i(t) = \sum_{\tau=1}^{p} a_i(\tau) s_i(t-\tau) + \eta_i(t) \tag{1}$$

where $\eta_i(t)$ and $a_i(\tau)$ denote a Gaussian noise term and a set of AR model parameters, respectively. As an alternative, the state variable $s_i(t)$ may formally be modelled by a convolution of a fixed function $\alpha_i(t)$ and a Gaussian noise process $\eta_i(t)$, similar to a *Wold representation* [6]:

$$s_i(t) = \sum_{\tau=-\infty}^{\infty} \alpha_i(t-\tau) \eta_i(\tau) \tag{2}$$

In a linear state space model, the observation equation would typically be given by an *instantaneous* observation model:

$$y_i(t) = c s_i(t) + \epsilon_i(t) \tag{3}$$

where $\epsilon_i(t)$ and c denote another Gaussian noise term and an observation parameter, respectively. However, in a situation with multipath propagation, this equation should be replaced by a *convolutive* observation model:

$$y_i(t) = \sum_{k=0}^{K} c(k) s_i(t-\tau_k) + \epsilon_i(t) \tag{4}$$

where K denotes the number of indirect paths; the direct path corresponds to $k = 0$. The numbers τ_k and $c(k)$ denote sets of delay times and of summation weights, respectively. As a convention, we set $\tau_0 = 0$ and $c(0) = 1$. Eq. (4) describes the data $y_i(t)$ by a weighted sum of delayed copies of a latent signal $s_i(t)$, corresponding to a multipath propagation situation. This observation equation can also be applied to general observation processes with temporally extended impulse responses.

Cepstral F statistics

The methodology of cepstral analysis has been introduced by Bogert *et al.* in 1963 [1]; it has found many applications to fields like seismic data processing, speech analysis, gear-box fault diagnosis and others. We will now review a variant of cepstral analysis which embeds the estimation of the cepstrum into a statistical F test [6]. For this purpose, we start from Eqs. (2) and (4), which we transform into frequency domain; then the power spectrum of $y_i(t)$ becomes

$$|Y_i(f)|^2 = |A_i(f)|^2 \, |C(f)|^2 \, |H_i(f)|^2 + |E_i(f)|^2 \tag{5}$$

where $Y_i(f)$, $A_i(f)$, $C(f)$, $H_i(f)$ and $E_i(f)$ denote the Fourier transforms of $y_i(t)$, $\alpha_i(t)$, $c(t)$, $\eta_i(t)$ and $\epsilon_i(t)$, respectively. In particular, we will have

$$|C(f)|^2 = \sum_{k=0}^{K} \sum_{l=0}^{K} c(l)c(k) \cos\big(2\pi f(\tau_l - \tau_k)/T\big) \qquad (6)$$

We shall now make the assumption $|E_i(f)|^2 \ll |Y_i(f)|^2$ and neglect $|E_i(f)|^2$ in Eq. (5). Then, after taking logarithms and rearranging, we obtain

$$\log \frac{|Y_i(f)|^2}{|A_i(f)|^2} = \log |C(f)|^2 + \log |H_i(f)|^2 \qquad (7)$$

Bogert *et al.* suggested to regard the logarithm of a power spectrum as a pseudo time series (of length $T/2$) and to estimate the power spectrum of this pseudo time series, to be called "*cepstrum*", in order to determine the time delays τ_k and their differences $\tau_l - \tau_k$, see Eq. (6):

$$Q_i(d) = \frac{1}{\sqrt{T}} \sum_{f=0}^{T/2-1} \log \frac{|Y_i(f)|^2}{|A_i(f)|^2} \exp(2\pi j f d/T) \qquad (8)$$

The frequency variable d of the cepstrum is called "*quefrency*"; it has the unit of time and is interpreted as a variable denoting "delay".

The (inverse) Fourier transforms of $\log |C(f)|^2$ and $\log |H_i(f)|^2$ shall be denoted by $S(d)$ and $V_i(d)$, respectively; then from Eq. (7) we have

$$Q_i(d) = S(d) + V_i(d) \qquad (9)$$

In this equation, following Shumway *et al.* [6], we regard $S(d)$ as a common signal, to be retrieved, and $V_i(d)$ as a noise term at each individual hydrophone. While $\log |Y_i(f)|^2$ can be computed from the data, $\log |A_i(f)|^2$ is assumed to be a relatively smooth function that can be modelled by a cubic spline function [6]. The classical approach to detecting a common signal in a set of stationarily correlated time series is given by testing the null hypothesis $S(d) = 0$ by a cepstral ANOVA, i.e., by an F test. The F statistic is given by [5]

$$F_{2,2(N-1)}(d) = (N-1) \frac{N|\bar{Q}(d)|^2}{\sum_{i=1}^{N} |Q_i(d) - \bar{Q}(d)|^2} \qquad (10)$$

where $\bar{Q}(d)$ denotes the mean of the cepstra of the individual hydrophones:

$$\bar{Q}(d) = \frac{1}{N} \sum_{i=1}^{N} Q_i(d)$$

The denominator in Eq. (10) represents the *cepstral noise component*. The subscripts in $F_{2,2(N-1)}(d)$ refer to 2 degrees of freedom (df) in the numerator and $2(N-1)$ degrees of freedom in the denominator.

The cepstral F statistic provides a level of significance that can be compared with a chosen false alarm rate (probability of type I error), say 0.01; then the corresponding significance threshold can be computed from the inverse of the cumulative F distribution function, at the given pair of df values [5]. By this approach, the plain estimation of the cepstrum is converted into a proper statistical test.

Since in many cases we have data from only a single sensor, i.e., $N = 1$, we suggest to create a multivariate time series by subsampling combined with "data recycling": as an example, if a subsampling factor of $N = 2$ is chosen, then the original univariate time series

$$y(1), y(2), y(3), y(4), y(5), y(6), y(7), y(8), \ldots$$

would be replaced by the bivariate time series

$$\begin{pmatrix} y(1) \\ y(2) \end{pmatrix}, \begin{pmatrix} y(3) \\ y(4) \end{pmatrix}, \begin{pmatrix} y(5) \\ y(6) \end{pmatrix}, \begin{pmatrix} y(7) \\ y(8) \end{pmatrix}, \ldots$$

and this easily generalises for higher subsampling factors.

Application to a real data set

We apply the methodology of analysis by cepstral F statistic, as described above, to data recorded during the REPMUS 2021 campaign, where REPMUS stands for "Robotic Experimentation and Prototyping Using Maritime Unmanned Systems". The data have been recorded by a stationary bottom node (type "NEREUS MK II", designed and built by our department) in the Atlantic Ocean, close to the city of Setubal, Portugal, at position 38°21′50.4″ N, 8°58′14.25″ W. The node was situated at a water depth of about 89 metres, with its top hydrophone 1.4 metres above the sea floor. A fisherman's ship was sailing at a speed of about 5 metres/second, approximately along a straight line, in Western direction. The hydrophone recorded the water sound of the ship for a time interval of 40 minutes duration, with the *point of closest approach* (CPA) occurring after about 20 minutes. The distance at CPA was about 140 metres. Original sampling frequency is 36 kHz.

In Fig. 1 the geometrical situation during the recording at the time of CPA is illustrated. The ship ("sound source")

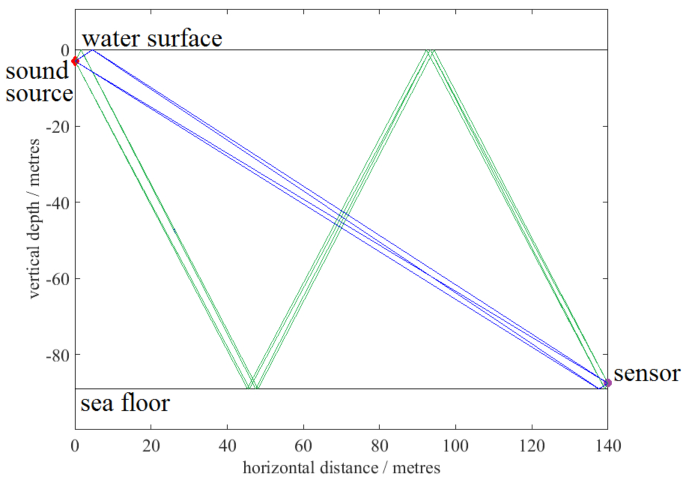

Figure 1: Geometrical situation during the recording at the point of closest approach; two groups of paths are shown: $k = 0, 1, 2, 3$ (blue) and $k = 4, 5, 6, 7$ (green)

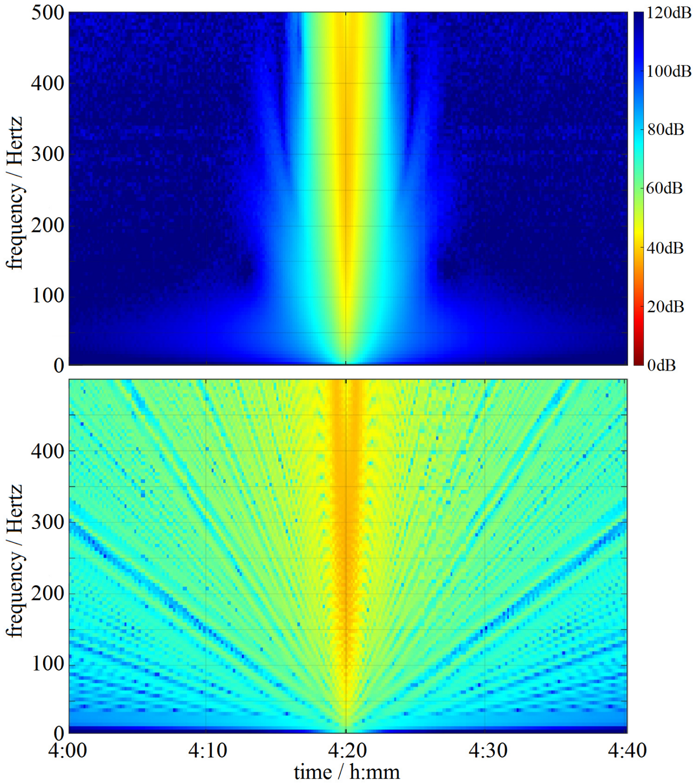

Figure 2: Theoretical spectrogram (transmission loss) for a complete passage of a ship, for the given geometric situation (see Fig. 1, either without sea floor (upper panel) or with sea floor (lower panel), computed by the OASES software.

is shown on the left side (red dot); it is assumed to be located 3 metres below the surface. The hydrophone ("sensor") is shown on the right side (purple dot). A subset of 8 paths from source to sensor is displayed, corresponding to $k = 0, 1, \ldots, 7$.

In the figure, it can be seen that due to the small distance between the sound source and the surface, as well as the small distance between the sensor and the sea floor, the chosen subset of paths forms two groups of paths with similar lengths within each group, one shown with blue lines, the other with green lines; by increasing K beyond the chosen value of 7 more such groups could have been displayed.

In Fig. 2 the theoretical spectrogram (transmission loss) for a complete passage of a ship, for the geometric situation which corresponds to the given real data, is shown, either without sea floor (upper panel) or with sea floor (lower panel); these results have been computed by the "Ocean Acoustic and Seismic Exploration Synthesis" (OASES) software [4], assuming a sea floor composed of sand and a realistic sound velocity profile.

In the case without sea floor only two paths contribute to the interference pattern, the direct path and the path with reflection at the surface, such that a classical Lloyd's mirror effect results. In contrast to this, in the case with sea floor a much more complicated interference pattern results, with strong interference effects persisting through the entire time interval of 40 minutes duration.

In Fig. 3 (upper panel) the actual spectrogram of the real hydroacoustic time series is shown, based on non-parametric estimation by DFT. In the figure horizontal lines can be seen, corresponding to the hydroacoustic signature of the passing ship; however, the spectrogram is dominated by the interference pattern resulting from multipath propagation in shallow water. This pattern consists of striations with approximately hyperbolic shape, centered around the CPA. In the power spectra of short data segments these striations will produce spurious peaks, thereby impeding analysis of the hydroacoustic signature.

In Fig. 4 the cepstral F statistic $F_{2,2(N-1)}(d)$ is displayed

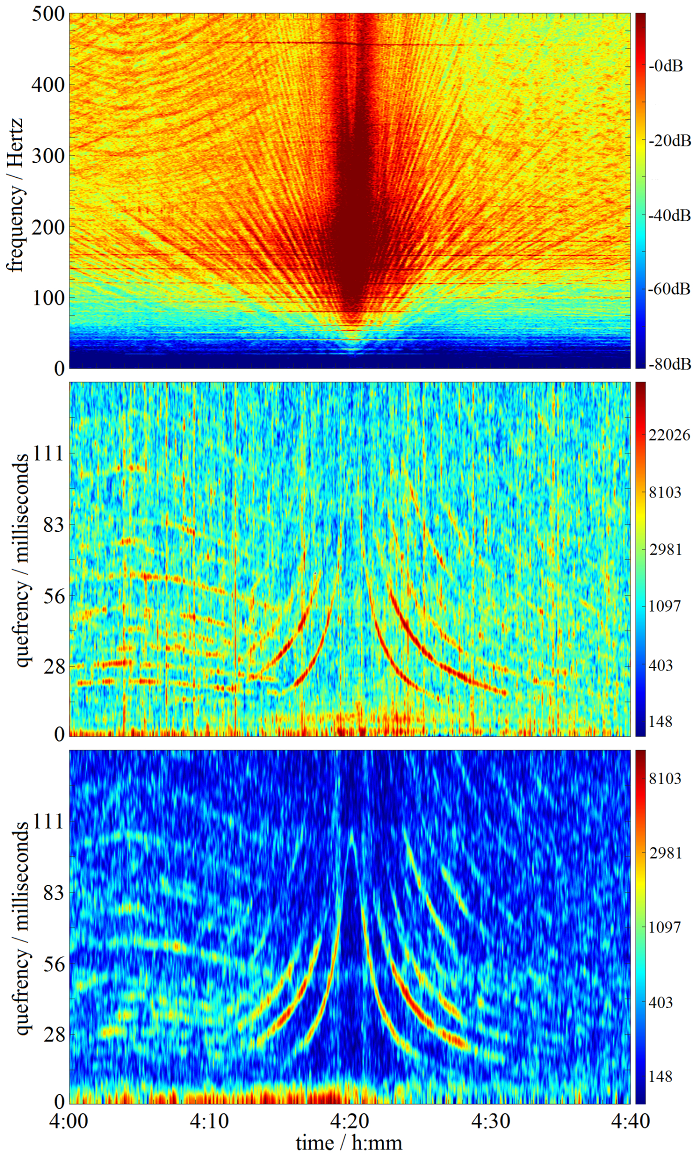

Figure 3: Spectrogram (upper panel), F statistic cepstrogram using a subsampling factor of $N = 10$ (middle panel) and F statistic cepstrogram using a subsampling factor of $N = 20$ (lower panel) of a hydroacoustic time series of a passing ship, with CPA at time 4:20; the spectrogram was estimated by a moving window of 0.44 seconds length; the cepstrograms were estimated by a moving window of 2.0 seconds length.

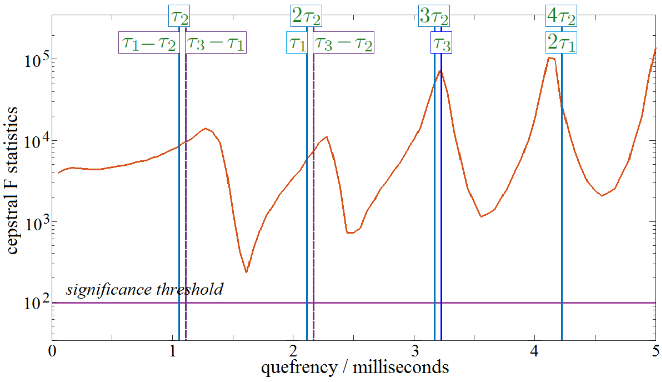

Figure 4: Cepstral F statistic $F_{2,2(N-1)}(d)$ for a time series of 6 seconds length at CPA (red) and corresponding F test significance threshold (purple); vertical lines denote the delay times τ_k, $k = 1, 2, 3$, and their multiples and differences, as indicated. The subsampling factor has been chosen as $N = 2$.

as a function of the quefrency d, for a segment of 6 seconds length, selected from the complete hydroacoustic time series at the approximate time of the CPA. In the figure it can be seen that for the quefrency interval displayed, the statistic is always above the significance threshold. Four maxima of the statistic are visible; these correspond approximately to the delay times τ_k, $k = 1, 2, 3$, and to their multiples and differences, as indicated in the figure. As can be seen, these delay times are below 5 milliseconds; they belong to the first group of paths displayed in Fig. 1 (blue lines). The maxima corresponding to the second group (green lines) can also be detected; for such detection it is preferable to choose a time point different from the time of CPA (compare Fig. 3, middle and lower panel).

Similar as with the power spectrum, also for the cepstral F statistic it is possible to compute and display its time-varying behaviour, thereby obtaining the *(F statistic) cepstrogram*. In the middle and lower panels of Fig. 3 the cepstrogram of the complete hydroacoustic time series is shown, for subsampling factors of $N = 10$ and $N = 20$. In these panels, a pattern of bell-shaped curves is visible, centered around the CPA; the pattern is better visible for higher subsampling factor. Obviously, these curves correspond to the passage of the ship; similar patterns have been obtained by Ferguson *et al.* [2] in a simulation study.

As a consequence of choosing larger values for the subsampling factors, the first group of paths (blue lines in Fig. 1), with delay times below 5 milliseconds, cannot be resolved in the cepstrograms. The pronounced curves which are visible in the cepstrograms, result from the second group of of paths (green lines), and from the multiples of the corresponding delay times. In contrast, for the results shown in Fig. 4, we have chosen a subsampling factor of $N = 2$, which was sufficient for resolving the delay times of the first group of paths.

Conclusion and outlook

In this paper we have applied cepstral analysis, namely its variant based on the F test statistic, to a hydroacoustic data set, recorded during the passage of a ship in a multipath environment. It is obvious that hydroacoustic multipath propagation represents a situation of convolutive observation, such that there would arise the need for *deconvolution*.

We have demonstrated that cepstral analysis can detect the delay times of low-order multipaths, as they usually arise in shallow water. For typical geometries, with sound sources and hydrophones very close to the water surface and/or the sea floor, comparatively high sampling rates may be needed, in order to discriminate paths with small differences of length.

For the purpose of modelling hydroacoustic multipath propagation, geometric information will be helpful, if it is available. Alternatively, it may be possible to estimate some geometrical parameters from the data; this task would represent an *inverse problem* which in this case could also be called a *blind deconvolution* problem.

As a final comment we would like to state that a convolutive observation does not necessarily represent a disadvantage; rather it may even provide additional information for the characterisation of the sound source. In future work we intend to apply the principle of *optimal information fusion* to hydroacoustic data from multipath environments.

References

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*Note that the words "cepstrum", "quefrency", "alanysis" and "saphe" are NOT typographical errors; they were deliberately created by the authors of [1].